Feature Based Random Forest Nurse Care Activity Recognition Using Accelerometer Data

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ABSTRACT

The *The 2nd Nurse Care Activity Recognition Challenge Using Lab and Field Data* addresses the important issue about care and the need for assistance systems in the nursing profession like automatic documentation systems. Data of 12 different care activities were recorded with an accelerometer attached to the right arm of the nurses. Both, laboratory and field data were taken into account. The task was to classify each activity based on the accelerometer data. We participated as team *Gudetama* in the challenge. We trained a Random Forest classifier and achieved an accuracy of 61.11% on our internal test set.

CCS CONCEPTS

• **Human-centered computing → Ubiquitous and mobile computing**; • **Computing methodologies → Neural networks**.

KEYWORDS

Nurse Activity Recognition Challenge, supervised learning, Random Forest, IMU, activity recognition, classification

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1 INTRODUCTION

Socio-demographic developments of industrialised countries like Germany are characterised by low birth rates and an increasing life expectancy. As the number of people reaching old age increases, it is likely that there will be more people who will be in need of a long-term care in the future. It has been shown that life expectancy is increasing both inside and outside the care sector: Between 2001 and 2015, the life expectancy of men without care increased by 3.1% from 73.6 years to 75.9 years in Germany. However, the relative duration of care increased by 29.4% from 1.7 to 2.2 years [5].
There is already a discrepancy between the supply of caregivers and the demand for caregivers and this discrepancy will probably grow. In an international comparison, Germany has the worst patient-to-caregiver ratio in Europe already, with measurable effects on patient mortality rates and the stress experience of caregivers [1, 9].
The nursing profession is often characterised by stress due insufficient time to take care of the patients as well as documentation. Therefore it is important to lessen the burden on nursing staff. Documentation of work processes requires a lot of time that can be better used for patient care. Therefore automatic recognition of nursing actions is a valuable step to support nurses in their work and reduce stress.
Smartphones are suitable sensor platforms for the detection of activities, as they are widely used and equipped with a variety of sensors. They consist of an inertial measurement unit (IMU), which usually includes an accelerometer and a gyroscope. The accelerometer measures the acceleration of the device. As care activities often follow the same pattern, the data can be used to automatically identify and document

the work steps.
The *The 2nd Nurse Care Activity Recognition Challenge Using Lab and Field Data* addresses the important issue about care and the need for assistance systems in the field of nursing. The challenge includes real field data, which represents everyday care that is better than laboratory data, but also includes challenges in recognition.
We, the team *Gudetama*, used a Random Forest approach to classify the 12 different care activities. The article at hand is structured as follows: The following section gives a brief overview of the state of the art. In Section 3 the used dataset, the pre-processing pipeline, the feature selection and the used algorithm are described. Thereafter, the results are shown and discussed in the subsequent section. At the end we draw conclusions and give some prospects for future work.

2 STATE OF THE ART

In recent years, extensive work in the field of sensor-based human activity recognition has been carried out, which is attracting an increasing attention in various application areas. This section introduces the state of the art related to mobile human activity recognition in general [4, 13, 15] and in particular activity recognition of medical professionals using accelerometers [3, 11, 16].
Different approaches have been used to recognise human activities. Bao and Intille [4] found that the k-Nearest-Neighbour (kNN) approach is outperformed by Decision Trees using mean, energy, frequency-domain entropy, and correlation features. For the recognition of everyday activities, an accuracy of 84% was achieved by using five biaxial accelerometers attached to hip, wrist, arm, ankle, and thigh. The same sensor modalities were used by Mannini and Sabitini [15], who used classifiers based on Hidden Markov Models (HMMs). An accuracy of 98.4% was achieved in the classification of 20 human physical activities. Other approaches focused on the recognition of activities by smartphone data from a smartphone positioned in the participant's pocket. Therefore, Kwapisz et al. [13] used the following features: average acceleration, standard deviation, average absolute difference, average resultant acceleration, time between peaks and binned distribution. Bahle et al. [3] showed that a smartphone carried by a nurse in the coat pocket can be used to document patient care activities. A variety of sensors such as acceleration, gyroscope, compass, WiFi and sound were used. The approach of Naya et al. [16] focused on combining location information and accelerometers to classify 13 different nursing activities.
The first nurse care activity recognition challenge [14] aimed to classify seven care-related activities performed by eight subjects in a controlled laboratory environment using accelerometers, a motion capture system and indoor location sensors. For the challenge Kadir et al. [12] proposed a method using an ensemble of kNN classifiers that use data from motion captured skeleton and meditag sensors. An accuracy of 80.2% was achieved on the test set. Cao et al. [6] applied a spatial-temporal graph Convolutional Network to process the 3D motion capture data. The deep learning approach achieved an accuracy of 64.6%.

3 MATERIALS AND METHODS

Dataset

The Heiseikai data, nurse care activity dataset [2, 10] has been acquired by using a 3-axis accelerometer of a smartphone with a sampling rate of $60Hz$. During data acquisition, the smartphone was attached to the right arm of a professional nurse. With a total of 47 participants the data was collected in both laboratory and in real life settings. For the challenge a part of the dataset was released. The laboratory data of two and the field data of six participants were provided for the training and the final test set is comprised of the field data of three participants who were not included in the training set.
The care activities of the dataset can be divided into three main categories: help in mobility, assistance in transfer, and position change. Each of these can further be divided into four sub-categories, resulting in a total of 12 activities which can be categorised as guide (from the front), partial assistance, walker, wheelchair, all assistance, partial assistance from front, side and back, to supine position, to right lying position, to left lying position, lower body lifting and horizontal movement.

Table 1: The result of the balance analysis on the training set. # samples is the number of samples in the label file which have corresponding data. The duration denotes the execution time for each activity (day-hours:minutes:seconds).

Label	Lab		Field	
	# Samples	Duration	# Samples	Duration
1	33	0-00:32:39	118	0-01:11:09
2	35	0-00:34:43	7	0-00:06:22
3	35	0-00:34:47	29	0-03:55:33
4	35	0-00:36:56	342	1-10:27:26
5	35	0-00:34:32	423	2-03:00:02
6	35	0-00:34:36	14	0-00:35:09
7	35	0-00:34:33	182	0-06:34:30
8	35	0-00:34:39	4	0-00:05:30
9	35	0-00:34:35	62	0-00:36:04
10	35	0-00:34:32	72	0-00:36:23
11	35	0-00:34:34	1	0-00:00:59
12	39	0-00:38:33	223	0-02:26:09

Pre-processing

The challenge test set consist of field data from users who were not included in the training set and we decided to use only field data for our validation and test set, but we merged laboratory and field data to our new training set. To overcome the class imbalance samples were deleted or copied. Table 1 shows the the number of samples and the time for each label. The training set was balanced to a threshold of 30 samples per class for each environment. The validation set was balanced with a threshold of 5 samples per class and the test set with a threshold of 3 samples per class. If the timestamps *start* and *finish* were equal for one activity in the label file, $59s$ were added to *finish*. For some samples in the label file no values could be found. Those were excluded. Overall, there are 720 training samples, 120 validation samples and 36 test samples after balancing. Due to the decision to include field data only in the validation and test set and the small sample size cross-validation was not applicable. The next step was to compute the magnitude (1) of the sensor values to remove the influence of the orientation of the smartphone.

$$magnitude = \sqrt{x^2 + y^2 + z^2} \tag{1}$$

The magnitude was filtered by a second order low-pass filter with a cut-off frequency of $15Hz$. From the filtered magnitude seven features described in the next section were derived.

Feature Selection

We considered various time and frequency domain features. In total, seven features were extracted from the filtered magnitude of acceleration, whereof six were time domain features (mean, standard deviation, signal energy, signal magnitude area, autocorrelation, root mean square) and one was a frequency domain feature (spectral entropy). Each feature has a specific benefit, e.g. it has been shown that the use of mean and signal energy leads to accurate recognition of postures and activities [4].

The mean value of the magnitude of a time series segment x indicates the amplitude of the segment [17].

$$M(x) = \overline{x} = (n - (m-1))^{-1} \cdot \sum_{i=m}^{n} x_i \tag{2}$$

Using the signal energy (E), activities with high energy can be distinguished from activities with low energy. The signal energy is calculated as follows:

$$E(x) = \sum_{i=m}^{n} x_i^2 \tag{3}$$

The standard deviation is used to measure the stability of the measured values of a time series segment x.

$$SD(x) = \sqrt{(n - (m-1) - 1)^{-1} \cdot \sum_{i=m}^{n} (x_i - \overline{x})} \tag{4}$$

Root mean square (RMS) reflects the amplitude of signals and can be used as follows:

$$RMS(x) = \sqrt{\frac{\sum_{i=m}^{n} x_i^2}{(n - (m-1))}} \tag{5}$$

By using the signal magnitude area (SMA) it is possible to distinguish dynamic from static activities.

$$SMA(x) = (n - (m-1))^{-1} \cdot \sum_{i=m}^{n} |x_i| \tag{6}$$

Autocorrelation (AC) describes the correlation of a signal with itself at an earlier point in time. It is used to detect periodic activities with repeating patterns in time series'.

$$AC(x_{i,i+j}, x_{n,n+m}) = \mathbf{E}[x_{i,i+j} \times \overline{x}_{n,n+m}] \tag{7}$$

with $x_{i,i+j}$ and $x_{n,n+m}$ are intervals of the same signal, i, j, m, $n \leq$ signal length, and $\mathbf{E}$ the expected value. An efficient way to compute the AC is by using fast Fourier Transformation [19].

Furthermore, we extract the spectral entropy (SE) as a feature in the frequency domain [18]. This is suitable to distinguish activities or movements with similar energy values. The SE is computed by the following equations

$$P(FFT(x)) = FFT(x)\overline{FFT(x)} \tag{8}$$

$P(\circ)$ is called the power spectrum, x the signal, FFT Fast Fourier Transformation and $\overline{FFT(x)}$ denotes the complex conjugation. The next equation normalises the power spectrum

$$C \sum_{x=x_i}^{x_n} P(FFT(x_i)) = 1 \tag{9}$$

where C is a constant chosen, so that the equation holds. The x_i are frequencies of the signal. The next step sums the spectral entropy for a given frequency range

$$S(x_i, x_n) = \sum_{x=x_i}^{x_n} P(FFT(x_i) \log\left(\frac{1}{P(FFT(x_i))}\right) \tag{10}$$

where x_i is defined as before. In the last step S is divided by the logarithm of the number of frequency components N.

$$S(x_i, x_n) = \frac{S(x_i, x_n)}{log(N(x_i, x_n))} \tag{11}$$

for this research the whole signal, i.e. sample, is used as input.

One final sample is a 9-dimensional vector, because the features AC and SE are complex values.

Algorithm

For classifying the activities a Random Forest classifier was used [8]. Random Forests are ensemble classifiers and based on the idea that the combination of simple weak classifiers outperforms one simple weak classifier. A Random Forest consist of a certain number of Decision Trees and the classification is based on a majority vote of all trees. The individual trees are built with a bootstrap aggregating or bagging strategy, i.e. for building each tree a randomly selected subset of the training set is used. Since a Random Forest is an ensemble of Decision Trees methods for optimising Decision Trees like pruning can be applied. Moreover, the criterion for splitting a node while building the trees can be specified as well.

4 RESULTS

The used Random Forest classifier had 44 Decision Trees with a maximum depth of 14. Each tree learner was trained with a randomly selected subset of size 193 with replacement. The splitting criterion was the *entropy* criterion. The validation accuracy was 78.33% and the accuracy on the internal test set was 61.11%. Table 2 shows the confusion matrix of our internal test set. The classes with the best true classifications are in the activity group help in mobility 2 (partial assistance) and 3 (walker), in the group assistance in transfer 6 (partial assistance from the front) and 8 (partial assistance from the back), and in the group position change 11 (lower body lifting). The classes with the most false classifications are in the activity group assistance in transfer: 5 (all assistance) and 7 (partial assistance from the side).

5 DISCUSSION

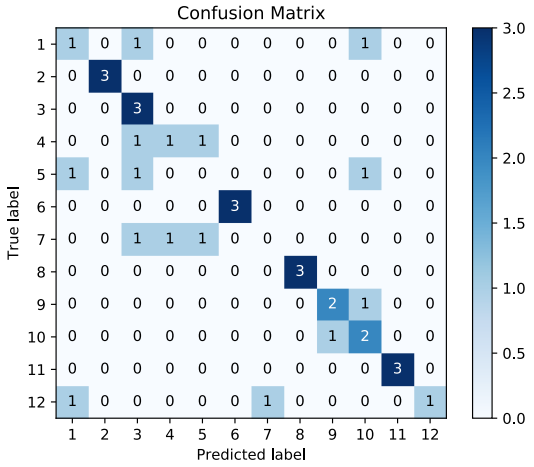
The classifier achieves the best results for the underrepresented classes, because our balancing strategy led to having the same samples in the training and in the test set. That is the case for the classes 2,3,6,8, and 11. The reason for the lower performance for the overrepresented classes (4,5,7 and 12) with a lot of values is similar. For these classes, all three sets are disjoint.

For classes where field and laboratory data where balanced only by deleting (1,4,5,6,7, and 12), we get the best impression on the performance of the classifier on the challenge test set, because the classes do not contain copies which could bias the result.

6 CONCLUSION & FUTURE WORK

In this article, we developed a machine learning approach to classify different nursing activities. Therefore, the test set was balanced to a threshold of 30 samples per class for both environments (laboratory and field). The same holds for the validation set with a threshold of 5 samples per class and

Table 2: The confusion matrix for the internal test set. The classes with the most false classifications are in the group assistance in transfer: 5 and 7. The classes with the best true classifications are in the group of help in mobility 2 and 3, assistance in transfer 6 and 8 , and position change 11.



the test set with a threshold of 3 samples per class. After computing the magnitude the samples were low-pass filtered and 7 features were derived. On this basis, we trained a Random Forest classifier with 44 Decision Trees and a maximum depth of 14. The achieved accuracy on our internal test set was 61.11%. Further improvements in our approach could lie in a better composition of training and test set. For example, more sophisticated over- and undersampling techniques could be used to conquer the challenge of imbalanced data. Simple deleting of data of over-represented classes (undersampling) could result in the loss of crucial variances, which represent the activities. In future work, controlled undersampling techniques like cluster-based undersampling approaches [20] should be used. The SMOTE [7] algorithm could be used as oversampling techniques for minority classes.

If more data would be available, the individual activities and movements of the hand/arm during these activities should be examined more closely. In this case, a feature adjustment tailored to nursing activities would be useful. The orientation of the sensor and therefore the direction of the acceleration should also be considered. We removed the information on the direction of acceleration by calculating the magnitude, as it was not clear whether the orientation of the smartphone on the arm was always the same. This explains, for example, the overlaps in classes 9 and 10, which were each classified one as the other.

Another point to be considered is the mix of laboratory and field data in the training set, while the test and validation sets

consisted of field data only. It is well known that laboratory data often differ from field data, because the standardised procedures are adapted to the local conditions and the actions may be influenced by external factors. This leads to a further challenge that has to be solved for the real application for the documentation of nursing activities.

Regarding the documentation privacy and surveillance are of great concern. On the one hand, using smartphones for documentation of nursing activities gives the full control of the data to the nurse. It is difficult for the employer to monitor Key-Performance-Indicators like activities per hour or served patients per hour. A monitoring would increase the pressure on the already stressed nurses. On the other hand, context information are missing. One important information is who currently is taken care of. The documentation of the treatments, assistance and condition is focused on the patient and not on the nurse. Adding context information would enable an assisted more holistic documentation.

The recognition result for the testing dataset will be presented in the summary paper of the challenge [2].

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REFERENCES

[1] Linda H Aiken, Walter Sermeus, Koen Van den Heede, Douglas M Sloane, Reinhard Busse, Martin McKee, Luk Bruyneel, Anne Marie Rafferty, Peter Griffiths, Maria Teresa Moreno-Casbas, et al. 2012. Patient safety, satisfaction, and quality of hospital care: cross sectional surveys of nurses and patients in 12 countries in Europe and the United States. *Bmj* 344 (2012), e1717.

[2] Sayeda Shamma Alia, Paula Lago, Adachi Kohei, Tahera Hossain, Hiroki Goto, Tsuyoshi Okita, and Sozo Inoue. 2020. Summary of the 2nd Nurse Care Activity Recognition Challenge Using Lab and Field Data. In *Proceddings of HASCA*.

[3] Gernot Bahle, Agnes Gruenerbl, Paul Lukowicz, Enrico Bignotti, Mattia Zeni, and Fausto Giunchiglia. 2014. Recognizing hospital care activities with a coat pocket worn smartphone. In *6th International Conference on Mobile Computing, Applications and Services*. IEEE, 175–181.

[4] Ling Bao and Stephen S Intille. 2004. Activity recognition from user-annotated acceleration data. In *International conference on pervasive computing*. Springer, 1–17.

[5] Statistisches Bundesamt. 2018. Datenreport 2018. Ein Sozialbericht für die Bundesrepublik Deutschland. (2018).

[6] Xin Cao, Wataru Kudo, Chihiro Ito, Masaki Shuzo, and Eisaku Maeda. 2019. Activity Recognition Using ST-GCN with 3D Motion Data. In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers* (London, United Kingdom) *(UbiComp/ISWC '19 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 689–692.

[7] Nitesh V Chawla, Kevin W Bowyer, Lawrence O Hall, and W Philip Kegelmeyer. 2002. SMOTE: synthetic minority over-sampling technique. *Journal of artificial intelligence research* 16 (2002), 321–357.

[8] Tin Kam Ho. 1995. Random Decision Forests. In *Proceedings of the Third International Conference on Document Analysis and Recognition (Volume 1) - Volume 1 (ICDAR '95)*. IEEE Computer Society, USA, 278.

[9] Ulrike Höhmann, Manuela Lautenschläger, and Laura Schwarz. 2016. Belastungen im Pflegeberuf: Bedingungsfaktoren, Folgen und Desiderate. *Pflege-Report* (2016), 73–89.

[10] Sozo Inoue, Sayeda Shamma Alia, and Paula Lago. 2020. *Nurse Care Activities Datasets: In laboratory and in real field.* https://doi.org/10.21227/jem3-ap07 Accessed, Jul. 17, 2020.

[11] Sozo Inoue, Naonori Ueda, Yasunobu Nohara, and Naoki Nakashima. 2016. Recognizing and understanding nursing activities for a whole day with a big dataset. *Journal of Information Processing* 24, 6 (2016), 853–866.

[12] Md. Eusha Kadir, Pritom Saha Akash, Sadia Sharmin, Amin Ahsan Ali, and Mohammad Shoyaib. 2019. Can a simple approach identify complex nurse care activity? *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers* (2019).

[13] Jennifer R Kwapisz, Gary M Weiss, and Samuel A Moore. 2011. Activity recognition using cell phone accelerometers. *ACM SigKDD Explorations Newsletter* 12, 2 (2011), 74–82.

[14] Paula Lago, François Charpillet, Sozo Inoue, Sayeda Shamma Alia, Shingo Takeda, Tittaya Mairittha, Nattaya Mairittha, Farina Faiz, Yusuke Nishimura, Kohei Adachi, and Tsuyoshi Okita. 2019. Nurse care activity recognition challenge. In *Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers - UbiComp/ISWC '19*. ACM Press.

[15] Andrea Mannini and Angelo Maria Sabatini. 2010. Machine learning methods for classifying human physical activity from on-body accelerometers. *Sensors* 10, 2 (2010), 1154–1175.

[16] Futoshi Naya, Ren Ohmura, Fusako Takayanagi, Haruo Noma, and Kiyoshi Kogure. 2006. Workers' routine activity recognition using body movements and location information. In *2006 10th IEEE international symposium on wearable computers*. IEEE, 105–108.

[17] Nishkam Ravi, Nikhil Dandekar, Preetham Mysore, and Michael L. Littman. 2005. Activity Recognition from Accelerometer Data. In *AAAI*.

[18] H. Viertiö-Oja, V. Maja, M. Särkelä, P. Talja, N. Tenkanen, H. Tolvanen-Laakso, M. Paloheimo, A. Vakkuri, A. Yli-Hankala, and P. Meriläinen. 2004. Description of the Entropy™ algorithm as applied in the Datex-Ohmeda S/5™ Entropy Module. *Acta Anaesthesiologica Scandinavica* 48, 2 (2004), 154–161.

[19] Norbert Wiener. 1964. *Time Series.* M.I.T. Press, Cambridge, Masachusetts. 42 pages.

[20] Show-Jane Yen and Yue-Shi Lee. 2006. Under-sampling approaches for improving prediction of the minority class in an imbalanced dataset. In *Intelligent Control and Automation*. Springer, 731–740.

APPENDIX

Table 3: Hardware Overview

Part	Lenovo NeXtScale System	Dell XPS 15	Dell Latitude 5401
CPU	2x Intel Xeon E5-2650 12x2.2GHz	Intel Core i9 8x2.4GHz	Intel Core i7 6x2.6GHz
GPU 1	NVIDIA Tesla P100 16GB HBM2	Intel UHD Graphics 630	Intel UHD Graphics 630
GPU 2	*not available*	NVIDIA Geforce GTX 1650 4GB GDDR6	*not available*
GPU 3	*not available*	NVIDIA TITAN V 12GB HBM2 (Thunderbolt 3)	*not available*
RAM	256GB DDR4 2400MHz	32GB DDR4 2677MHz	16GB DDR4 2677MHz
Mass Storage	1TB HDD	1TB NVMe SSD	512GB NVMe SSD

Table 4: Additional Information

Sensor Modalities & Features	Accelerometer, Signal Energy, Mean, Autocorrelation, Standard Deviation, Root Mean Square, Signal Magnitude Area, and Spectral Entropy
Programming Language & Libaries	Python 3.6, Pandas, Numpy, Scikit-Learn
Window Size & Post Processing	*not available*
Training & Testing Time	$0.075s$ & $0.675s$